

Jade Shi

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Summary

Full Stack **.NET Developer** with a **University of Melbourne IT** master's degree and **AWS Certified Cloud Practitioner** credential. Production experience **designing, building and maintaining secure, scalable software** using **C# .NET 6+**, **ASP.NET Core WebAPI**, **React/Next.js**, **SQL Server**, **Azure**, **Docker** and **GitHub Actions CI/CD**. Experienced in **payment processing**, **event-driven architecture**, idempotency, Outbox pattern, automated testing, performance testing, observability and production deployment. Strong focus on clean, maintainable code and solving real business problems through practical engineering.

Technical Skills

Backend: C#, .NET Core, ASP.NET Core, WebAPI, REST APIs, EF Core, SQL Server, Azure SQL

Frontend: React, Next.js, TypeScript, Razor Pages, HTML, CSS

Architecture & Integration: Microservices, layered architecture, API integration, event-driven architecture, Outbox Pattern, RabbitMQ/MassTransit, Stripe

Cloud & DevOps: Azure, Azure App Service, Azure Static Web Apps, Docker, GitHub Actions CI/CD, AWS (Cloud Practitioner)

Testing & Quality: xUnit, Moq, unit testing, integration testing, API error-path testing, code coverage, clean code

Security & Reliability: JWT/RBAC, resource ownership checks, idempotency, audit logging, structured logging, correlation IDs

Performance & Observability: k6, latency percentiles, throughput, error-rate tracking, troubleshooting

Tools & Delivery: GitHub, Agile/Scrum, GitHub Copilot, Cursor, stakeholder communication

Project Experience

NDIS Food Ordering & Payment Platform | Live Site *Melbourne, Australia* *Feb 2025 – Jul 2025*
ASP.NET Core, Next.js, React, SQL Server, Stripe, JWT, Azure, RabbitMQ/MassTransit, Docker, GitHub Actions

- Designed and built a production-style food ordering and payment platform covering User, Service, Order and Payment workflows across frontend and backend; developed **4 backend service domains** with **ASP.NET Core WebAPI**, exposing **11 REST API endpoints** with layered architecture, DTO contracts and SQL Server persistence.
- Implemented **multi-layer idempotency** for order and payment flows (client-generated keys, DB constraints, Stripe idempotency keys) — preventing duplicate charges during retries, page refreshes and network failures, demonstrating expertise in **transactional business solutions** and **payment processing**.
- Decoupled payment creation from Order API using **event-driven architecture** (RabbitMQ/MassTransit + **Outbox Pattern**); improved **fault tolerance** and failure isolation across services.
- Secured all 11 endpoints with **JWT/RBAC** and resource ownership checks; built **36 unit and integration tests (89–90% line coverage)** covering happy paths, idempotency behaviour and external-service failure paths.
- Validated deployed API performance with **k6** (20 virtual users, 3,062 requests, **0% error rate**); measured **p95 latency at 148ms** and added **structured logging** with correlation IDs for request tracing and troubleshooting.
- Implemented a **RabbitMQ/MassTransit event-driven** CreateOrder flow alongside the deployed API, evaluating trade-offs in **reliability**, **failure isolation** and operational complexity — demonstrating architectural decision-making across synchronous and async messaging patterns.
- Delivered via **GitHub Actions CI/CD (47 successful workflow runs)**; deployed to **Azure App Service** and **Azure Static Web Apps** with environment-based configuration; used **Docker** for repeatable local development.

Hotel Late Check-in Payment Pre-authorisation System | Freelance Delivery — Live in Production *Melbourne, Australia* *Feb 2026 – Present*
ASP.NET Core, Razor Pages, Stripe, SMTP, PDF Generation, Azure, GitHub Actions

- Sole developer contracted by a Melbourne motel to replace an after-hours phone service; system is **currently live and in active use** by real guests and staff, saving **~\$1,200/year** in operating cost — demonstrating end-to-end ownership from requirements through to **production delivery**.
- Built a secure guest self-service workflow covering guest details, photo ID upload, terms acceptance and e-signature; created an auditable digital record for each check-in replacing phone-based coordination.
- Integrated **Stripe pre-authorisation** for a **\$100 security bond** with staff-controlled capture and release states; separated test/live credentials to reduce credential leakage risk.
- Built a **JWT-secured staff admin workflow** with SMTP email notifications and PDF summary generation; delivered via **GitHub Actions CI/CD** (14 successful workflow runs) and Azure deployment.
- Collaborated directly with the motel stakeholder to clarify front-desk requirements, translate operational constraints into technical decisions and iterate on deployment needs — demonstrating **stakeholder communication** and the ability to find **practical solutions to challenging problems**.

Professional Experience

NIO

Software QA team member

China
2021 – 2022

- Embedded in an **Agile delivery team** at a large-scale automotive tech company, collaborating with **Product Owners, developers and QA** across sprint cycles; investigated defects, communicated root causes to developers and supported release verification — building firsthand understanding of **scope control, stakeholder communication** and **production-quality delivery**.

Education & Certifications

Master of Information Technology, University of Melbourne

Melbourne, Australia

AWS Certified Cloud Practitioner